

Timing the Market: A Skewed Perspective

How much market timing matters is a property of the market, not a universal law

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1. Introduction

In an era of rapid artificial-intelligence development and the infrastructure build-out accompanying it, equity markets sit in an advantageous position to capture whatever gains that phenomenon eventually produces. Whether the resulting inflation in market values is warranted is a separate question, and not one this paper attempts to settle. What it does raise, for young adults and first-time investors in particular, is a more immediate one: when is the best time to enter the market? How do we use entries to dampen volatility? Is it better to wait for a substantial drop, or simply to begin now at near all-time highs?

The conventional answer arrives as a proverb:

“Time in the market beats timing the market.”

Versions of this claim recur across interviews, panels, letters to shareholders, and popular investment writing. Among those associated with it:

- Michael LeBoeuf — “Timing the market is for losers. Time IN the market will get you to the winner’s circle, and you’ll sleep better at night.”
Quoted in his 2002 book The Millionaire in You (LeBoeuf 2002).
- Ken Fisher — “Time in the market beats timing the market – almost always.”
From his 2018 column in USA Today (Fisher).
- Peter Lynch — “Far more money has been lost by investors preparing for corrections, or trying to anticipate corrections, than has been lost in corrections themselves.”
From an essay in Worth Magazine (Lynch).
- Warren Buffett — “The stock market is a device for transferring money from the impatient to the patient.”
From Berkshire Hathaway’s 1992 annual shareholder letter (Buffett).

A common denominator among these figures is that all four are American, and made their reputations investing in American equities. For those who live in, or allocate to, other markets, the question is how far the claim actually travels. It may be a generalisation drawn from a single market that was unusually well positioned to capture the gains of the post-Cold-War period and the technology era that followed (Jorion and Goetzmann 1999).

This paper argues that the notion of “time in market” is a property of the results of the American market rather than a universal law.

The main measurement of this paper is the “timing dispersion,” which is the spread of the 10,000 investors in each market, randomly simulated to choose entry points in each market. It serves as a function to explore whether the timing of entry matters to the final wealth (more on Section 4.5).

The idea is illustrated with four investors following different recognisable strategies and shown what the variations meant in practice.

2. Prior Work

Various papers have explored the idea of time in the market over timing the market, and each informs the design below.

2.1 The “worst market timer” genre

Carlson’s Bob invests only at four peaks between 1972 and 2007, yet retires with \$1.1m against \$2.3m from annual averaging (Carlson 2014). Schwab’s perfect timer beats immediate investment by just \$15,522 over twenty years (Schwab Center for Financial Research).

2.2 Cross-country studies of timing

Estrada draws 160,278 daily returns from fifteen markets: missing the best ten days cuts terminal wealth 50.8% on average (Estrada 2008). Estrada tests foresight rather than a followable rule.

2.3 Buy-the-dip as a tested heuristic

AQR tests 196 dip strategies on the S&P 500 and finds average Sharpe below passive, with 20% thresholds generating almost no signals (Cao et al. 2025). Bonini, Shohfi and Simaan report start-date sensitivity and estimation risk (Bonini et al. 2024). Both report averages on US data; neither maps the distribution of outcomes elsewhere.

2.4 The behavioural gap

Dichev’s dollar-weighted returns fall systematically below buy-and-hold across nearly all major international markets (Dichev 2007), though Keswani and Stolin show the gap is not robust (Keswani and Stolin 2008). Aggregate flows are a residual: they record what investors collectively did, not what one specified, replicable behaviour would have produced.

2.5 The gap this paper fills

No study combines a pre-specified mechanical trigger, a long-run multi-market panel, and the full distribution of outcomes rather than averages. That combination is what makes the underlying claim falsifiable — whether “time in the market beats timing the market” survives outside the American drift regime that produced it.

3. The Design

The first measurement that this paper employs is timing dispersion. In each market, 10,000 simulated investors contribute the same total sum over the same twenty-five years, differing only in the dates on which they buy. The timing dispersion is the “score”.

To further illustrate, four hypothetical investors are inserted, each receiving the same income on identical schedules, to be allocated to the same market. These investors differ only in their strategies, and the timing in which they put their investment to work. This illustrates what the dispersion figure means for a rational investor, and makes the investigation easier to relate to observed market behaviour.

4. Data and Method

4.1 Markets

The starting point is the largest national equity markets by total market capitalisation as of December 2000, subject to being continuously investable by a US-dollar investor for the full sample period. Six were retained from that set, chosen to span the range of drift while keeping the panel tractable.

Market	Index	Role in the argument
United States	MSCI USA	Baseline — norm for market comparisons
Japan	MSCI Japan	Low drift. The canonical counterexample.
United Kingdom	MSCI UK	Medium drift, dividend-heavy, minimal tech.
Germany	MSCI Germany	Medium drift, cyclical.
Hong Kong	MSCI Hong Kong	High volatility, China exposure.
Canada	MSCI Canada	High US correlation, resource-heavy.

Table 1. Markets, indices, and the role each plays in the argument.

These markets have been selected due to their unique disposition to the US market. Canada is retained precisely because it is expected to behave much like the United States, acting as a semi-control sample.

The panel was kept to a tractable six; France and Switzerland could equally have been included, and were excluded on that ground rather than on the size criterion. China is excluded because mainland shares were not continuously investable by a foreign investor over the sample period, while Hong Kong offers a partial glimpse into its market. Markets also had to survive throughout the period. Russia, which became effectively worthless to foreign holders and was removed from MSCI indices in 2022, and Greece, which was reclassified out of developed-market status, are the obvious exclusions (Jorion and Goetzmann 1999).

4.2 Index data

All returns use MSCI country indices, denominated in US dollars, at monthly frequency. MSCI is used in preference to each country's headline index because it applies one methodology everywhere: each index covers roughly the same proportion of its home market's tradeable value, dividends are treated identically, and the series is already converted to dollars. Headline indices are not comparable in this way — the DAX includes reinvested dividends by construction while the FTSE 100 and Hang Seng do not, so placing them side by side compares bookkeeping as much as performance.

All returns are gross of dividend withholding tax. Withholding rates depend on the treaty between the investor's country of residence and the market they invest in, so any single set of rates would embed one particular investor's tax position into the results. Gross figures isolate market behaviour from tax residency and let readers apply their own rates.

MSCI country indices are constructed through a three-step process (heavily simplified):

1. It collects the listed securities in the market, screened for the equity universe minimum size requirement.
2. Then only free-float shares are counted for the companies within the requirement.
3. The companies are then ranked by their adjusted market cap and added to the index until the MSCI index covers 85% of all the free-float-adjusted market cap of the country. The target is a proportion instead of a fixed number of companies or value, allowing the index to represent the same share of its respective home market.

4.3 Period

The sample runs from December 2000 to December 2025, a full twenty-five years. The window was chosen for three main reasons:

4. It is the longest span of history available for all six markets at once. Freely accessible MSCI index data reaches further back for some countries than others, and the analysis requires every market to run over an identical window.
5. All six markets were developed, liquid, and open to foreign investors throughout.
6. Twenty-five years is adequate to contain full market cycles and roughly matches the horizon a young investor faces over their working accumulation years. The window covers four distinct regimes: the collapse of the technology boom, the global financial crisis, the pandemic, and the inflation shock that followed.

The start date is a constraint rather than a preference, but its effect is smaller than it first appears. December 2000 sits in the early stage of the technology unwind, so the experiment begins near a peak and walks into a multi-year decline. However the effects are minimal, because the contributions made near that peak are a vanishingly small part of the final outcome. An investor deploys \$100 in December 2000, but then makes another 299 contributions of the same size over the following twenty-five years, and by 2025 the US market had risen several times above its 2000 level. The handful of dollars invested at the top are swamped by the far larger sums that compounded through everything that came after.

The endpoint matters as much as the start, and probably more. Japan is the clearest case: the MSCI Japan passed its 1989 high in 2024 and has risen further since, so a version of this paper ending in 2013 would conclude that Japan was a disaster in which timing was decisive, while a version ending in 2025 concludes that it recovered and rewarded patience. December 2025 is used because it is the most recent complete year, but the choice is consequential rather than neutral, and it is not tested here.

4.4 Contributions and currency

Each investor receives \$100 USD to invest in the respective markets each month. The frequency and timing in which they utilise the capital is determined by their own strategies. Factors such as inflation and rising cost of living mean the purchasing power of \$100 USD decreases over time, but the amount stays identical to ease comparison.

Contributions, cash, and returns are measured in US dollars, as MSCI reports in USD, which avoids an additional currency conversion step.

Contributions not yet deployed are held in cash earning the US three-month Treasury bill rate (FRED series DTB3, month-end observations). Because the investor modelled here is a US-dollar allocator, idle capital sits in dollars and earns the dollar risk-free rate regardless of which market it is destined for.

4.5 The Central Measure

Timing dispersion is defined as the gap between the ninety-fifth and fifth percentile of final wealth, divided by the median, across ten thousand simulated investors who contribute the same total sum on randomly drawn dates.

Each of the 10,000 simulated investors is allocated the same total capital and deploys it in equal instalments across n randomly drawn month-ends, holding everything to December 2025. Capital and the number of deployments are held constant; only the choice of months differs between investors. The four investors in Section 4.6 can only deploy money they have already received. That constraint is deliberately removed here, because tying entry dates to how much capital had accumulated by each date would confound the very effect being measured. These investors are a measuring instrument rather than a model of a real saver.

The measurement is run twice. The first uses the real market history, so that investors differ only in when they bought. The second uses a blocked bootstrap, which resamples each market's own returns in twelve-month blocks to construct alternative histories with similar statistical characteristics. One such history is held fixed across all investors in a given simulation, and the exercise is repeated across many histories.

For example, if the dispersion were to be clustered, it would mean investing at different times brought a similar final result, meaning the timing of the investments didn't really matter. However, if the timing dispersion was spread, it would mean that entry timing played a decisive role in the final wealth of the investments.

This will help us understand whether timing is relevant to the final wealth generated in the tested markets.

4.6 The investors

The table below sets out each investor's rules in full.

	1. Non-Investor	2. DCA	3. Dip Buyer	4. Hype Chaser
Role	Floor / lower bound	Benchmark ("time in the market")	Waits for weakness	Waits for strength
Contributions received	\$100/mo	\$100/mo	\$100/mo	\$100/mo
Entry rule	Never enters equities	Deploy full contribution at every month-end	Deploy all accumulated cash when index closes $\geq 20\%$ below trailing ATH	Deploy all accumulated cash when trailing 6-mo total return $\geq +10\%$
Holds cash	Always	Never	Between qualifying dips	Between qualifying rallies

Table 2. Investor definitions. The DCA column is the benchmark against which all others are measured.

Here, the dollar-cost-averaging investor stands as a representative of "time in market". This investor attempts to fully deploy the allocated capital and maximise time in the market.

The two timing investors (dip buyer, hype chaser) are not arbitrary. One waits for weakness and one waits for strength, which between them cover the two most common retail framings of entry timing. Including both means the finding does not rest on a single chosen rule. The dip threshold is taken from Bonini, Shohfi and Simaan, who test three depths (-5% , -10% and -20%) against a passive benchmark and a classical dollar-cost-averaging benchmark (Bonini et al. 2024).

The hype threshold uses a six-month lookback, which sits inside the three-to-twelve-month range over which Moskowitz, Ooi and Pedersen (2012) document time-series momentum across equity indices and other assets. The $+10\%$ level is not taken from that paper; it was set so the rule responds to a visible run of strength rather than ordinary monthly movement, and was not optimised against the results.

4.7 What is reported

Each investor is measured on the same basis: identical contributions, identical period, and final wealth compared directly. Because three of the four hold cash while waiting, the paper also reports how often each was out of the market, how many times capital was deployed, and months holding cash. These figures are not adjustments to the comparison — time spent out of the market is precisely what the thesis concerns, so it is part of the result rather than a distortion of it.

5. The Six Markets

So, does the proverb hold in all six markets? First, how the six markets performed over the sample period.

Market	Annualised return (drift)	Annualised volatility	Final wealth under DCA
USA	8.84%	15.31%	\$166,203
Canada	8.36%	19.60%	\$109,497

Market	Annualised return (drift)	Annualised volatility	Final wealth under DCA
Germany	6.17%	23.10%	\$90,120
Hong Kong	6.08%	20.42%	\$71,924
UK	5.22%	16.49%	\$73,763
Japan	4.54%	15.31%	\$73,594

Table 3. Annualised return, volatility, and final wealth under DCA.

The table is ordered by decreasing drift, and final wealth under DCA broadly follows that order. Two features are worth drawing out before the analysis begins.

The first is the drift spread itself. The United States returned 8.84% a year against Japan's 4.54% — a gap that compounds into more than double the final wealth on identical contributions. This spread is what the rest of the paper conditions on. It is not, by itself, evidence about timing: showing that low-drift markets produced less wealth under DCA says nothing about whether a timing rule would have beaten DCA in those markets. That question is explored in Section 7.

The second is that annualised return does not fully determine the outcome for a monthly accumulator. The United Kingdom, on 5.22% drift, finished slightly ahead of Hong Kong on 6.08%. The reason is sequence: a single annualised figure describes the journey throughout 25 years, but an investor contributing every month is exposed to when the gains arrived relative to when their money went in. A market that rose late rewards a monthly accumulator more than one that rose early, even at identical average returns.

6. How much did timing matter to a passive investor?

Following on from the previous section, we now look at the timing dispersions pertaining to our simulated random 10,000 investors in the real market; the synthetic-market results are reported in the appendix.

To establish the number of entries made by the simulated investors, the experiment used 100 entries as the basis. This was derived from taking a round number similar to our dip buyer (approximately 74 average entries) and hype chaser (approximately 95 average entries) investors. The paper also employed a sensitivity analysis to investigate the effects of proportionality of the entries, which was as expected.

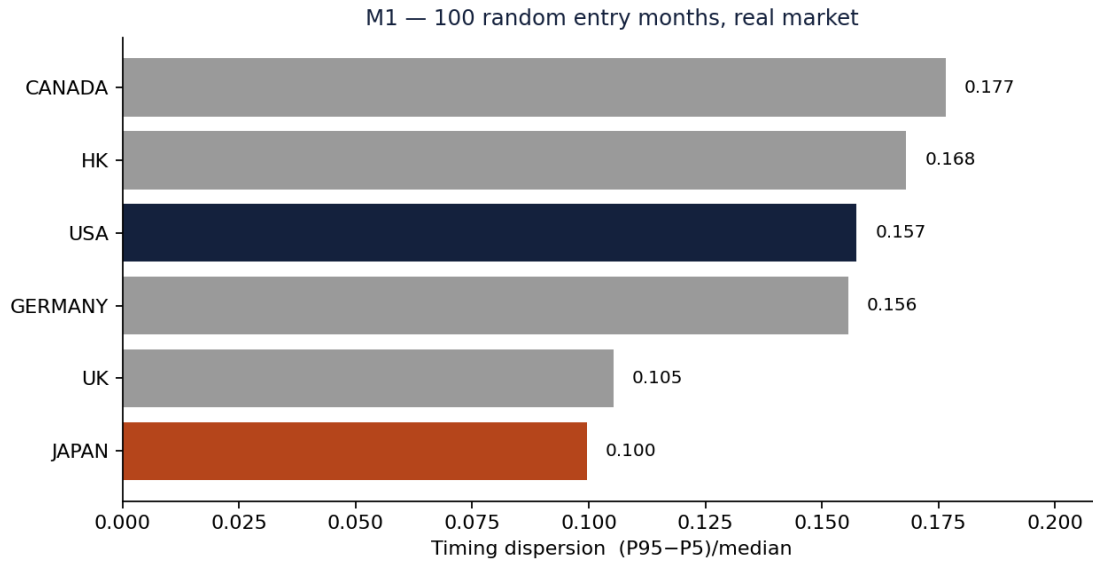


Figure 1. Timing dispersion at 100 random entry months, real market.

Across all six markets, entry timing mattered far less than expected. The variation that did occur did not follow volatility. The most volatile markets were not where timing dispersion was highest, and two markets with near-identical volatility (the US and Japan) showed very different sensitivity to entry date. Again, if anything predicts where timing mattered, it is drift ($r = 0.761$), not volatility ($r = 0.596$).

The two markets where random, passive timing mattered least (the lowest timing dispersion scores) are Japan and the UK, the two lowest-drift markets. The market one would assume should have the highest timing-sensitivity because of its volatility (Germany, 23%, the most volatile) sits mid-pack, while low-volatility USA sits right up with it.

These figures are small in absolute terms. To check whether the dispersion observed in each market was peculiar to the single history that happened to occur, the same measurement was repeated on synthetic histories generated by block bootstrap — holding one synthetic path fixed across all simulated investors, so that they varied only in entry date, and repeating across many such paths. In every market the observed dispersion fell between the 32nd and 66th percentile of the synthetic distribution: comfortably inside the range a market with these characteristics would ordinarily produce. The synthetic medians themselves tracked drift at $r = 0.96$, more strongly than the real-market figures, so the synthetic test reinforces the drift finding rather than complicating it. Full results are given in the appendix.

Entry timing was the smallest of the levers. What dominated was exposure to the market itself: the market the investor chose, and the sequence of returns that market happened to deliver over their holding period. The passive date of entry, the variable investors agonise over most, moved outcomes least.

7. Could a deliberate timer do better?

In the section above, we established that random entries with no specific pattern barely mattered. But a random investor is not what the proverb warns against — it warns against people deliberately trying to time. So now we test actual rules: buy weakness (dip buyer), buy strength (hype chaser), against the benchmark of just staying invested (DCA).

As a reminder, a dip buyer waits for a 20% fall; the hype chaser waits for a 6-month +10% run.

Market	Annualised return	Dip Buyer vs DCA	Hype Chaser vs DCA
USA	8.84%	-12.6%	-2.2%
Canada	8.36%	-2.8%	-2.4%
Germany	6.17%	+2.7%	-2.6%
Hong Kong	6.08%	-3.4%	-3.7%
UK	5.22%	+3.8%	+0.7%
Japan	4.54%	+2.3%	+1.2%

Table 4. Results of dip buyer and hype chaser against a DCA investor.

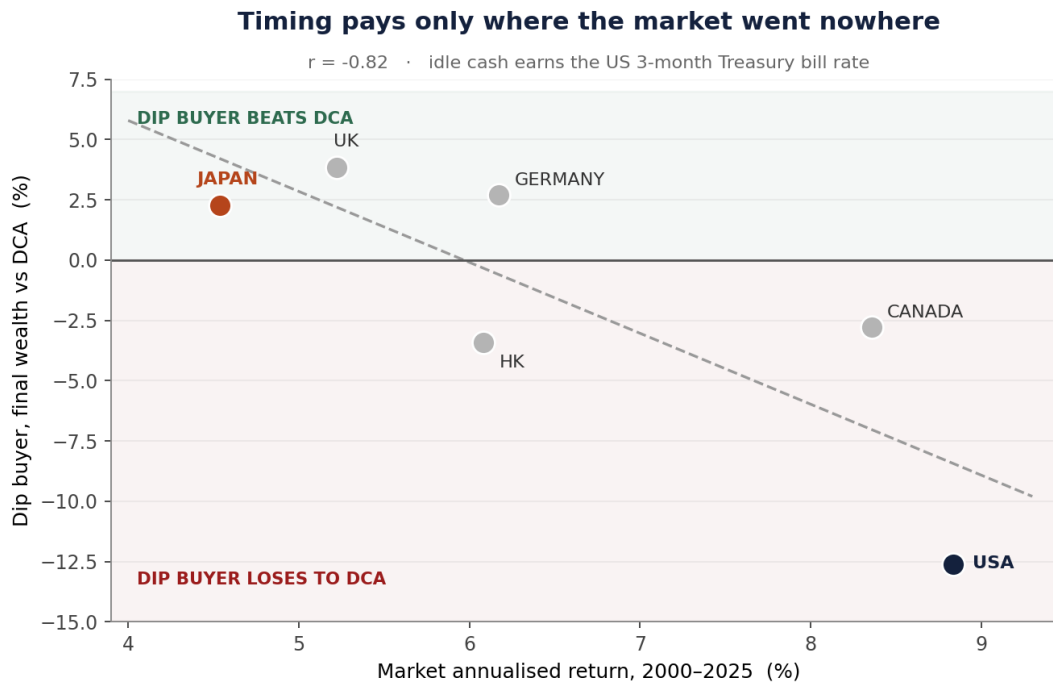


Figure 2. Dip-buyer advantage against market drift. Full results in the appendix.

As seen, the dip buyer beat DCA in Japan, the UK and Germany, the low-drift markets. It lost in the US and Canada, the high-drift ones. It also lost in Hong Kong, which at 6.08% has lower drift than Germany's 6.17% — the one market that does not fit the pattern, and the point sitting visibly below the fitted line

above. Across all six markets the correlation between dip-buyer advantage and annualised return is $r = -0.82$ ($n = 6$, descriptive); excluding Hong Kong it is -0.85 .

Dip buyers in the US were in the 0th percentile, beating none of the 10,000 simulated investors. Waiting for a dip in a rising market was not helpful. On the contrary, Japan (99.2) and UK (99.7) dip buyers were in the 99th percentile compared to their respective samples. Note that Germany is also at the 97.9th percentile, also beating DCA. See Appendix Figure A3.

The underlying reason is that in a rising market, time out of the market gets very expensive over time, as compounding takes over for those in the market. Anyone who waits pays a heavy drift penalty. In a sideways market like Japan and the UK, it is better to pick the timing to get into the market.

For hype chasers, the correlation between final wealth and the market's annualised return had the same sign as the dip buyer's but was materially weaker ($r = -0.60$ against -0.82), so the paper uses the dip buyer's figures to illustrate the mechanism. See Appendix Figures A4.

8. Drift as the hinge

In Section 6, we uncovered that for a passive and random investor, entry timing barely mattered, and the little correlation it had was with drift and not volatility.

In Section 7, we understood that for an investor with a specific strategy, timing actually matters more than time in market — especially in markets with low drift like the UK and Japan.

From the two sections, drift is the variable that matters.

The idea is that time out of the market is a cost, and that cost scales with drift. In a high-drift market, waiting for the perfect time to enter might cause an investor to lose out on incremental gains, and over time those minor increases lead to a massive effect with compounding. In a flat market, there are fewer incremental gains to miss out on, so waiting costs almost nothing, and avoiding bad stretches might net positive gains in return. Timing is not intrinsically good or bad; its value is set by the drift being forgone.

“Time in the market” dampens the effect of volatility, but an investor's final wealth is determined by drift, which is local and market specific. People often conflate the two. The proverb “time in the market beats timing the market” is really a statement about the exceptional drift in the US market rather than a control over one's strategic timing.

The proverb is true where drift is high, and the US has had the highest drift of any major market over the past quarter of a century. It is false where drift is low. It was formulated by American investors in the highest-drift major market of the era, then exported as universal law. It is not. It is an American artifact.

9. Limitations and Robustness

There are certain limitations to this paper's findings. We used six markets ($n = 6$), one realised history each, but those histories are correlated. So cross-market correlations ($r = 0.76$, $r = -0.82$) are descriptive, not tested. The paper draws a pattern for the reader to infer but does not establish a definitive trend.

Survivorship also cuts one way: the six markets all persisted, while those that did not — Russia, Greece — are absent, so the sample understates how severely exposure can be punished. The analysis was also run with idle cash earning nothing, as an alternative to the Treasury bill rate, with near-identical results — the finding is not sensitive to the cash assumption.

10. Conclusion

This paper set out to test a proverb. “Time in the market beats timing the market” is repeated as though it were a law of investing, but it was formulated by a small group of American investors describing an American market, and it has never been asked whether it travels.

Two independent tests were run across six developed markets over twenty-five years, and both pointed at the same variable. For a passive investor choosing entry dates at random, timing barely mattered anywhere, and the small variation that did occur tracked a market's drift, not its volatility. The United States and Japan carried near-identical volatility yet showed markedly different sensitivity to entry date, which rules out the intuition that turbulent markets are where timing counts. For a deliberate investor following an explicit rule, the pattern was sharper still: waiting for a twenty per cent drawdown beat simply staying invested in Japan, the United Kingdom and Germany, and lost in the United States and Canada. The correlation between a market's return and the penalty of timing the market was -0.82 . In the United States the dip buyer finished below all ten thousand randomly-timed investors; in Japan and the United Kingdom the same rule finished above ninety-nine per cent of them. One rule, one period, opposite verdicts.

The mechanism is that time spent out of the market is a cost, and that cost scales with drift. Timing is neither inherently wise nor inherently foolish. Its value is set almost entirely by the drift being given up while waiting.

This produces a distinction the proverb obscures. Spreading purchases across time is a defence against volatility, and that defence is arithmetic — it worked identically in all six markets. But what an investor actually ends up with, and whether patience was the right choice at all, is governed by drift, which is local. The two are routinely conflated. “Time in the market beats timing the market” is not really advice about patience; it is a description of what happens when a market rises relentlessly for a quarter of a century. The United States did exactly that, and the maxim was written there and exported everywhere else.

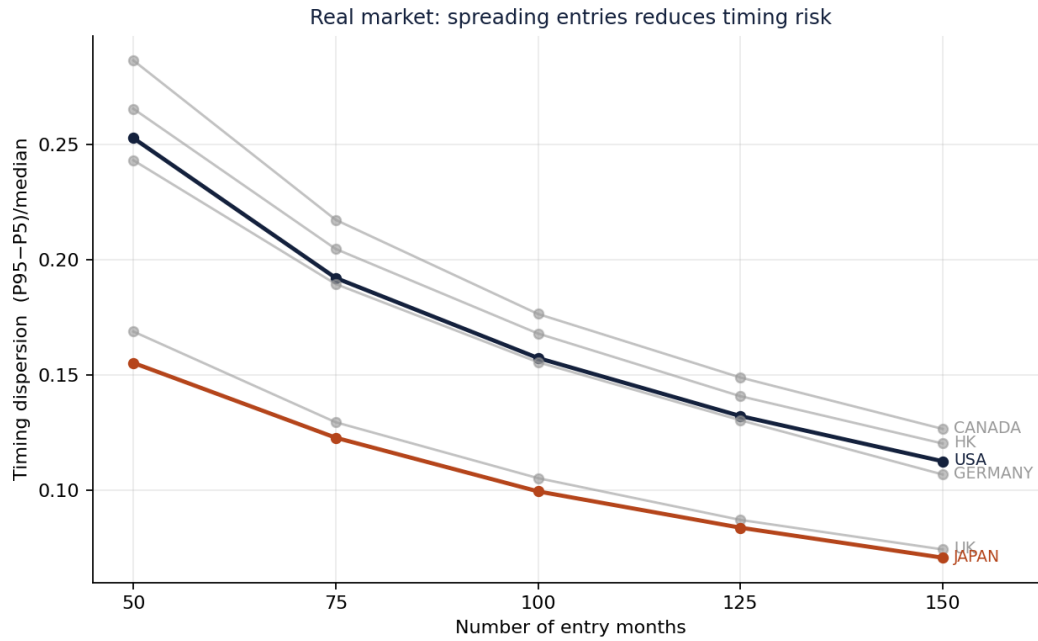
For any young investor deciding whether to enter at near all-time highs now or wait for a fall, the answer this evidence supports is narrower than the proverb but more useful. Entry timing is the smallest lever available. Across every market tested, the spread of outcomes caused by when money entered

was dwarfed by the spread caused by which market it entered and what that market did over the following twenty-five years. The question worth agonising over is not the month of entry. It is whether you have a long-term view, and whether that view coincides with a growth market. It could also be that you wish to invest in a local market that moves sideways — then perhaps one should time their entries.

Appendix

Table A1. Test 1 — simulated on the real market (timing dispersion)

Market	Annualised return	50 entries	75 entries	100 entries	125 entries	150 entries	Reduction 50→150
USA	8.84%	0.2528	0.1922	0.1574	0.1323	0.1127	55%
Canada	8.36%	0.2865	0.2174	0.1765	0.1490	0.1267	56%
Germany	6.17%	0.2432	0.1895	0.1556	0.1305	0.1069	56%
Hong Kong	6.08%	0.2655	0.2047	0.1681	0.1409	0.1204	55%
UK	5.22%	0.1689	0.1296	0.1053	0.0872	0.0744	56%
Japan	4.54%	0.1553	0.1229	0.0996	0.0839	0.0708	54%

Figure A1. Real market — spreading entries reduces timing risk

Measure	vs annualised return (drift)	vs annualised volatility
Real-market dispersion	0.761	0.596

Table A2. Correlation of real-market dispersion with drift and volatility.

Table A3. Test 2 — blocked bootstrap synthetic market (timing dispersion)

Market	50 entries	75 entries	100 entries	125 entries	150 entries	Reduction 50→150
USA	0.2799	0.2193	0.1742	0.1476	0.1241	56%
Canada	0.2731	0.2173	0.1726	0.1480	0.1205	56%
Germany	0.2470	0.1975	0.1508	0.1316	0.1082	56%
Hong Kong	0.2218	0.1720	0.1473	0.1218	0.0976	56%
UK	0.1934	0.1516	0.1281	0.1084	0.0896	54%
Japan	0.1738	0.1371	0.1086	0.0895	0.0779	55%

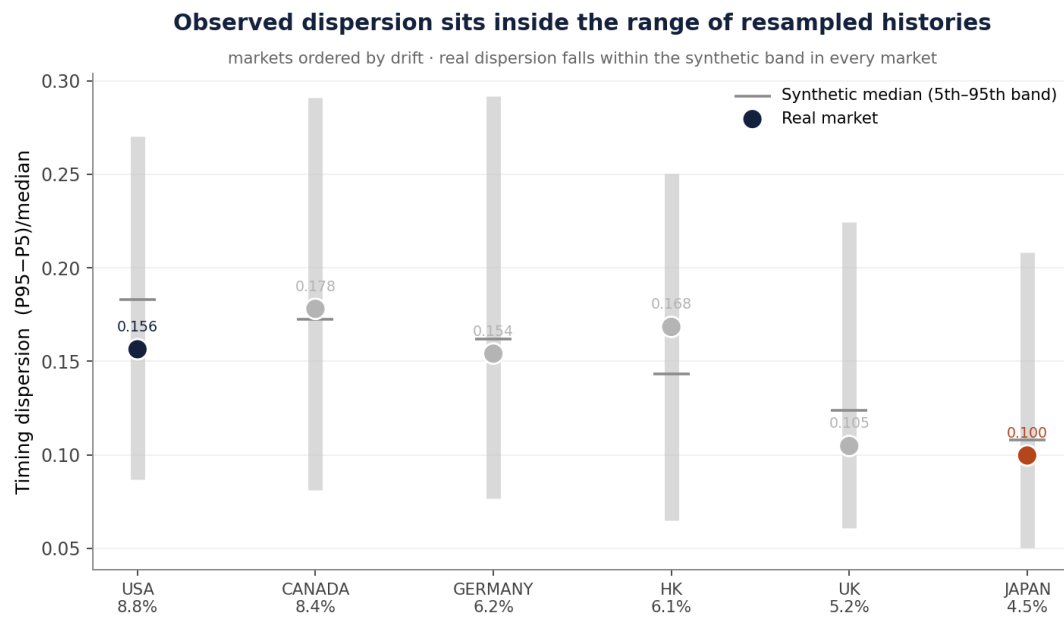
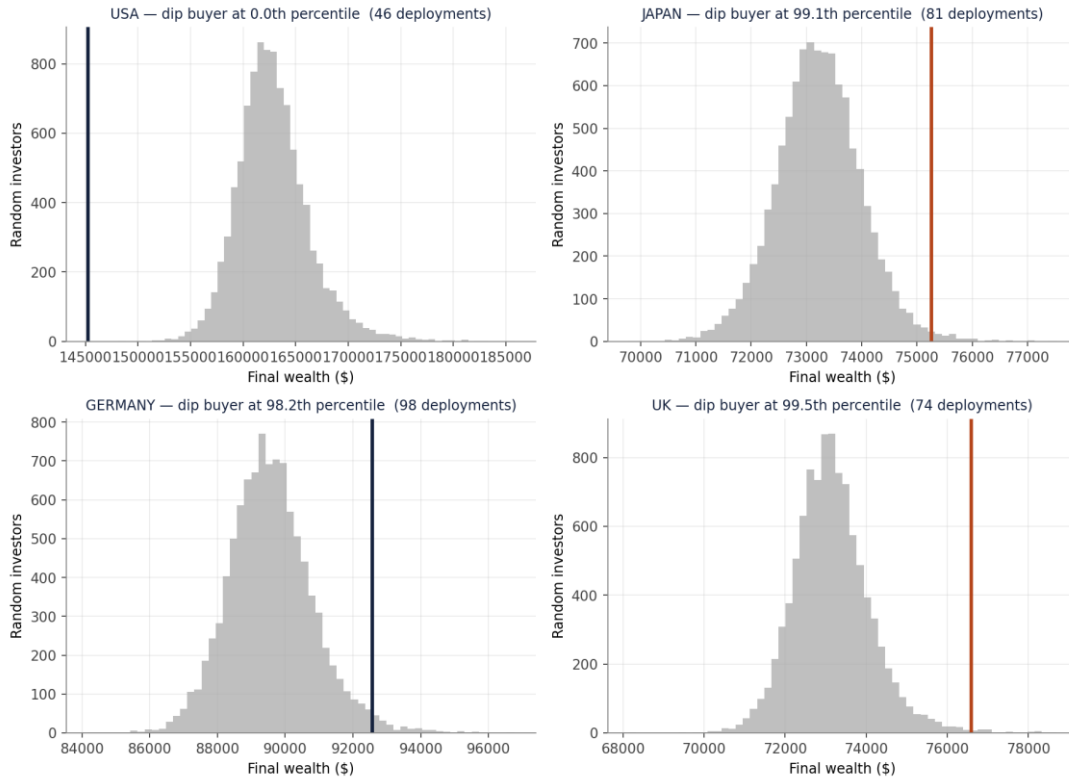
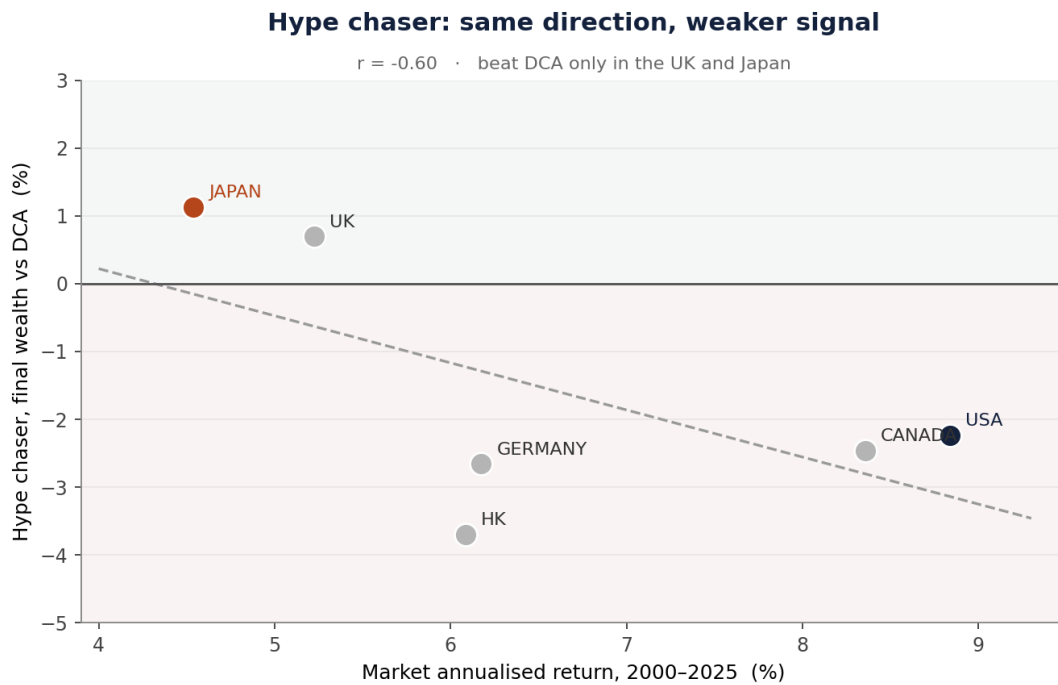
Figure A2. Observed dispersion against the range of resampled histories

Table A4. Test 3 — four strategies simulated in six markets

Market	Ann. return	Strategy	vs DCA (\$)	vs DCA (%)	Deploy events	Months holding cash	Final wealth
USA	8.84%	Non-Investor	-\$128,465	-77.3%	0	300	\$37,671
USA	8.84%	DCA	\$0	+0.0%	300	0	\$166,203
USA	8.84%	Dip Buyer -20%	-\$20,904	-12.6%	46	254	\$145,299
USA	8.84%	Hype Chaser	-\$3,671	-2.2%	92	208	\$162,532
Canada	8.36%	Non-Investor	-\$71,759	-65.5%	0	300	\$37,671
Canada	8.36%	DCA	\$0	+0.0%	300	0	\$109,497
Canada	8.36%	Dip Buyer -20%	-\$3,089	-2.8%	59	241	\$106,408
Canada	8.36%	Hype Chaser	-\$2,674	-2.5%	105	195	\$106,824
Germany	6.17%	Non-Investor	-\$52,381	-58.1%	0	300	\$37,671
Germany	6.17%	DCA	\$0	+0.0%	300	0	\$90,120
Germany	6.17%	Dip Buyer -20%	\$2,438	+2.7%	98	202	\$92,558
Germany	6.17%	Hype Chaser	-\$2,383	-2.6%	112	188	\$87,737
Hong Kong	6.08%	Non-Investor	-\$34,185	-47.5%	0	300	\$37,671
Hong Kong	6.08%	DCA	\$0	+0.0%	300	0	\$71,924
Hong Kong	6.08%	Dip Buyer -20%	-\$2,480	-3.4%	88	212	\$69,444
Hong Kong	6.08%	Hype Chaser	-\$2,638	-3.7%	95	205	\$69,286
UK	5.22%	Non-Investor	-\$36,024	-48.8%	0	300	\$37,671
UK	5.22%	DCA	\$0	+0.0%	300	0	\$73,763
UK	5.22%	Dip Buyer -20%	\$2,826	+3.8%	74	226	\$76,589
UK	5.22%	Hype Chaser	\$539	+0.7%	89	211	\$74,303
Japan	4.54%	Non-Investor	-\$35,855	-48.7%	0	300	\$37,671
Japan	4.54%	DCA	\$0	+0.0%	300	0	\$73,594
Japan	4.54%	Dip Buyer -20%	\$1,765	+2.3%	81	219	\$75,359
Japan	4.54%	Hype Chaser	\$868	+1.2%	75	225	\$74,462

Table A4. DCA rows are shaded as the benchmark. Green indicates the strategy beat DCA; red indicates it lost.

Figure A3. Dip buyers against their matched random benchmarks*Figure A3. USA, Japan, Germany and UK dip buyers against random investors deploying the same number of times.***Figure A4. Hype chaser advantage against market drift**

Bootstrap specification

Method. Synthetic market histories are constructed by fixed-length block bootstrap. Monthly log returns are divided into blocks of 12 consecutive months. Blocks are drawn at random with replacement and concatenated until the original series length is reached, then truncated to that length. The synthetic price path is $P^*_t = \exp(\text{cumulative sum of resampled returns})$, initialised at 1.0.

Block length. 12 months, chosen so that a full year of momentum through peaks and troughs is preserved within each block. Blocks rather than individual months are used because equity returns exhibit volatility clustering: drawing months independently would scatter a drawdown across the whole series rather than allowing it to occur as a run, making every market appear steadier than it was and understating dispersion (Politis and Romano 1994).

Draws. 10,000 simulated investors per market, per specification. A fixed synthetic path is generated for all 10,000 investors in the synthetic specification.

Entry months. Drawn without replacement. 100 entries in the headline specification, with a sweep at 50, 75, 100, 125 and 150.

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